

Arduino Morse Code Transmitter

Overview

This project implements a Morse code transmitter using an Arduino, LED, and buzzer. It converts serial text input into Morse signals using visual and audio feedback.

Hardware

- Arduino Uno
- LED on pin 11
- Buzzer on pin A1
- USB cable, breadboard, and jumper wires

Functionality

- Accepts text input via Serial Monitor.
- Each character is translated to Morse code.
- Dots: short LED and tone (300ms), Dashes: long (900ms).
- Delays separate symbols, characters, and words.

Code Highlights

- `dot()` and `dash()` control LED and buzzer timing.
- `morse(char)` maps characters to Morse using `switch-case`.
- Handles letters (A–Z) and digits (0–9).
- Uses `readString()` for full-line serial input.

Simulation

Tinkercad Circuit Online: <https://www.tinkercad.com/things/iz9SpQ6eUom-powerful-habbi>